

E069

Foundations of UI/UX

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A checkout page for a sneaker store founded in 2015 shows Pay Now, Save for Later, and Apply Coupon as three identical gray buttons. Analytics show many users stall on this screen. What change best applies visual hierarchy?

- A. Update the brand logo to feel more modern.
- B. Make all three buttons bright red.
- C. Add a paragraph explaining each button.
- D. Make Pay Now the one high-contrast primary button and style the others as secondary.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

On a long sign-up form built by a two-person agency, each label sits exactly halfway between the field above and the field below. Users often type their phone number into the email box. What is the best fix?

- A. Color every label a different color.
- B. Spread the fields farther apart evenly.
- C. Remove all labels and use only placeholder text.
- D. Move each label closer to its own field and add more space between field groups.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A regional hospital with a new parking garage wants a patient check-in app. The IT director says, “I know what patients want — big buttons and a dark theme.” No patients have been consulted. What should the design team do first?

- A. Survey drivers in the new garage about parking.
- B. Interview and observe patients at check-in to learn their needs.
- C. Build the design the IT director described.
- D. Make high-fidelity mockups in three color themes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A team designing a food-delivery app, whose office has a standing-desk policy, wants to agree on screen layout and navigation before choosing colors or fonts. Stakeholders meet on Thursday. What should the designer produce?

- A. Low-fidelity wireframes.
- B. A polished visual mockup.
- C. A fully coded front end.
- D. A brand style guide.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A designer needs users to click through a sign-up flow in a test to see where they get stuck. Which artifact fits?

- A. A user persona.
- B. A static wireframe.
- C. A clickable prototype.
- D. A high-fidelity static mockup.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

During a usability test in a newly painted lab, a participant struggles to find the Export button. The facilitator, who designed the page, says, "It's in the top right." The session runs 30 minutes. What went wrong?

- A. The session was too short.
- B. Nothing; helping keeps the participant comfortable.
- C. The facilitator led the participant, so the test cannot show whether users find Export on their own.
- D. The test should have asked whether the participant liked the page.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A team has budget for one round of usability testing on an early prototype. About how many representative users should they recruit to find most major problems?

- A. 1
- B. Every employee on the team
- C. At least 100, for statistical proof
- D. About 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A bank's website uses gray body text on a white background with a contrast ratio of 3.2:1. The designer, who has 12 years of experience, says it looks clean and readable. The text is normal size. Does it meet WCAG AA?

- A. No; all text must reach 7:1.
- B. Yes; 3:1 is enough for all text.
- C. No; normal text needs at least 4.5:1.
- D. Yes, because an experienced designer finds it readable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A headline is large text (24 px bold) with a 3.2:1 contrast ratio. Does it meet WCAG AA?

- A. Yes; large text needs 3:1.
- B. No; it needs 4.5:1.
- C. It depends only on the font family.
- D. No; headlines must reach 7:1.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A photo app used by 50,000 people uploads large files. After tapping Upload, the screen shows nothing for 8 seconds, and many users tap again, creating duplicates. The app has a five-star rating. Which fix best follows the visibility-of-status heuristic?

- A. Add a help page about uploads.
- B. Show a progress indicator right after the tap and disable the button while uploading.
- C. Nothing; the high rating shows users are satisfied.
- D. Delete duplicates later on the server.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A delivery drivers' app, launched in five cities, puts a tiny Mark Delivered button in the top-left corner, far from the driver's thumb. Drivers use it dozens of times a day. What does Fitts's law suggest?

- A. Make the button larger and place it within easy thumb reach near the bottom.
- B. Make it smaller to fit more features on screen.
- C. Change the button color only.
- D. Launch in more cities first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Two buttons are the same distance from the cursor. Button A is 20 px wide; Button B is 80 px wide. By Fitts's law, which is faster to click?

- A. They are the same, because distance is equal.
- B. Button B
- C. Whichever is a brighter color
- D. Button A

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A checkout page ~~for a sneaker store founded in 2015~~ shows Pay Now, Save for Later, and Apply Coupon as three identical gray buttons. Analytics show many users stall on this screen. What change best applies visual hierarchy?

Target word: emphasize the primary action **Decides:** Three equal buttons give no cue to the main action.

A	Camouflage	Brand age does not affect the button hierarchy.
B	Inverted operation	Equal emphasis again, just louder.
C	Adjacent category	Adds reading load instead of fixing emphasis.
D	Correct	Contrast and weight point users to the main action.

2. Answer D Concept 2

On a long sign-up form ~~built by a two-person agency~~, each label sits exactly halfway between the field above and the field below. Users often type their phone number into the email box. What is the best fix?

Target word: fix label-field grouping **Decides:** Equal spacing makes grouping unclear.

A	Adjacent category	Color does not fix the ambiguous spacing.
B	Inverted operation	Keeps labels equally distant from both fields.
C	Dropped element	Placeholders vanish while typing and hurt accessibility.
D	Correct	Proximity shows which label belongs to which field.

3. Answer B Concept 3

A regional hospital ~~with a new parking garage~~ wants a patient check-in app. The IT director says, “I know what patients want — big buttons and a dark theme.” No patients have been consulted. What should the design team do first?

Target word: first step in user-centered design **Decides:** No user data exists yet.

A	Camouflage	The garage is not the product.
B	Correct	Research comes before design decisions.
C	Confusion set	A stakeholder’s opinion is not user research.
D	Adjacent category	A design step that skips research.

4. Answer A Concept 4

A team designing a food-delivery app, ~~whose office has a standing desk policy~~, wants to agree on screen layout and navigation before choosing colors or fonts. ~~Stakeholders meet on Thursday~~. What should the designer produce?

Target word: right fidelity for structure decisions **Decides:** Layout first, before visual styling.

A	Correct	Show structure and layout without visual detail.
B	Inverted operation	Settles visual style before structure.
C	Absolute language	Far more effort than a layout decision needs.
D	Adjacent category	Defines colors and fonts, which the team wants to defer.

5. Answer C Concept 4 • transfer

A designer needs users to click through a sign-up flow in a test to see where they get stuck. Which artifact fits?

Target word: artifact for testing a flow **Decides:** It must be interactive.

A	Adjacent category	Describes users; it is not a design to test.
B	Inverted operation	Cannot be clicked through.
C	Correct	Prototypes are interactive, so users can move through the flow.
D	Confusion set	Detailed look, but still not interactive.

6. Answer C Concept 5

During a usability test ~~in a newly painted lab~~, a participant struggles to find the Export button. The facilitator, who designed the page, says, "It's in the top right." ~~The session runs 30 minutes.~~ What went wrong?

Target word: facilitator error **Decides:** The facilitator gave the answer.

A	Camouflage	Length is not the issue here.
B	Inverted operation	Help destroys the data.
C	Correct	Leading hides the usability problem the test should reveal.
D	Adjacent category	Opinion questions do not measure task success.

7. Answer D Concept 5 • transfer

A team has budget for one round of usability testing on an early prototype. About how many representative users should they recruit to find most major problems?

Target word: sample size for a formative round **Decides:** A small group finds most major issues; iterate after.

A	Dropped element	One user misses too many issues.
B	Confusion set	Team members are not representative users.
C	Absolute language	Formative testing seeks problems, not statistical significance.
D	Correct	Around five users per round surfaces most major issues.

8. Answer C Concept 6

A bank's website uses gray body text on a white background with a contrast ratio of 3.2:1. The designer, ~~who has 12 years of experience~~, says it looks clean and readable. The text is normal size. Does it meet WCAG AA?

Target word: contrast pass/fail **Decides:** Normal text needs 4.5:1.

A	Absolute language	7:1 is the stricter AAA level, not AA.
B	Confusion set	3:1 applies only to large text.
C	Correct	3.2:1 is below the AA threshold for body text.
D	Camouflage	The standard is a measured ratio, not an opinion.

9. Answer A Concept 6 • transfer

A headline is large text (24 px bold) with a 3.2:1 contrast ratio. Does it meet WCAG AA?

Target word: contrast for large text **Decides:** Large text needs 3:1.

A	Correct	3.2:1 clears the 3:1 bar.
B	Inverted operation	Applies the normal-text rule to large text.
C	Adjacent category	Size and weight matter; the contrast ratio decides.
D	Absolute language	7:1 is AAA, not AA.

10. Answer B Concept 7

A photo app ~~used by 50,000 people~~ uploads large files. After tapping Upload, the screen shows nothing for 8 seconds, and many users tap again, creating duplicates. ~~The app has a five-star rating.~~ Which fix best follows the visibility-of-status heuristic?

Target word: feedback during a wait **Decides:** Users get no sign the upload started.

A	Adjacent category	Users are not looking for help in the moment.
B	Correct	Tells users the system is working.
C	Camouflage	The rating does not decide this problem.
D	Dropped element	Treats the symptom, leaves users confused.

11. Answer A Concept 8

A delivery drivers' app, ~~launched in five cities,~~ puts a tiny Mark Delivered button in the top-left corner, far from the driver's thumb. Drivers use it dozens of times a day. What does Fitts's law suggest?

Target word: target size and distance **Decides:** Small, far, and frequently used.

A	Correct	Larger, closer targets are faster to hit.
B	Inverted operation	Smaller targets take longer to hit.
C	Adjacent category	Color helps visibility, not reach time.
D	Camouflage	Market size does not affect target design.

12. Answer B Concept 8 · transfer

Two buttons are the same distance from the cursor. Button A is 20 px wide; Button B is 80 px wide. By Fitts's law, which is faster to click?

Target word: compare by size **Decides:** Same distance; bigger target wins.

A	Dropped element	Ignores target size.
B	Correct	A larger target lowers selection time.
C	Confusion set	Color is not a term in Fitts's law.
D	Inverted operation	The smaller target is slower.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Visual hierarchy	Size, contrast, weight, and position tell users what matters most; the primary action should stand out and secondary ones recede.	Making every element equally bold so nothing stands out.	Users miss the main action on a screen.
2	Gestalt proximity	Items placed close together are read as a group; spacing shows which labels, fields, and controls belong together.	Fixing grouping problems with more color or lines instead of spacing.	Users pair a label with the wrong field.
3	User-centered design starts with research	Learn who the users are and what they need (interviews, observation) before designing; personas come from research, not guesses.	Designing from the team's own preferences or a stakeholder's opinion.	"We think users want..." with no data.
4	Fidelity: wireframe, mockup, prototype	Wireframes are low-fidelity layout and structure; mockups show static visual design; prototypes are interactive and used to test flows.	Calling a static high-detail image a prototype, or polishing visuals before structure is settled.	What the artifact must do: show layout, show look, or be clicked through.
5	Usability testing	Watch representative users try realistic tasks while thinking aloud; do not lead them; a handful of users (about five) per round finds most major issues.	Asking users whether they like it, or helping them when they get stuck.	A test plan or a facilitator's behavior.
6	Accessible contrast and color	WCAG AA needs 4.5:1 contrast for normal text and 3:1 for large text; never use color as the only signal.	Assuming a color that looks fine to the designer passes, or using red alone for errors.	A contrast ratio or a color-coded status.
7	Heuristic: visibility of system status	The system should always tell users what is happening through timely feedback, such as progress indicators and confirmations.	Treating a silent wait as fine because the task eventually finishes.	Users click twice or leave during a delay.
8	Fitts's law	Time to hit a target grows with distance and shrinks with target size, so frequent or important targets should be large and near.	Shrinking key buttons to save space.	Small, far-away, frequently used controls.

Course scope (public description): This course introduces learners to the foundational principles of user interface (UI) and user experience (UX) design within software engineering contexts. Learners explore visual design principles, user-centered design methodologies, wireframing and prototyping techniques, usability testing, and design-development collaboration. The course emphasizes creating functional, accessible interfaces that meet both user needs and technical requirements. This course is appropriate for learners in software engineering programs who need practical UI/UX skills for front-end development.